Feature Engineering Report

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Introduction

This report outlines the feature engineering procedures applied to the Ames Housing dataset. The purpose of these steps is to enhance data quality and prepare the dataset for machine learning models by creating meaningful features and converting categorical variables into numerical form.

Phase 1: Data Loading

The cleaned dataset (ames_clean.csv) was loaded successfully.

Initial Data Shape:

- Rows: 2930
- Columns: 82

Phase 2: Creating New Features

Three new features were engineered to enhance model performance:

1. **HouseAge:** Represents the age of the house at the time of sale.

   Formula: HouseAge = Yr Sold – Year Built

2. **RemodAge:** Represents the time since the last remodel or addition.

   Formula: RemodAge = Yr Sold – Year Remod/Add

3. **TotalSF:** Total square footage combining basement and floors.

   Formula: TotalSF = Total Bsmt SF + 1st Flr SF + 2nd Flr SF

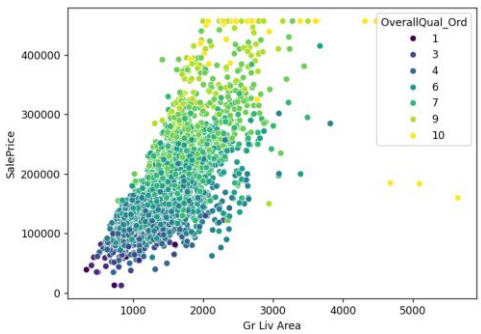
Phase 3: Encoding Categorical Variables

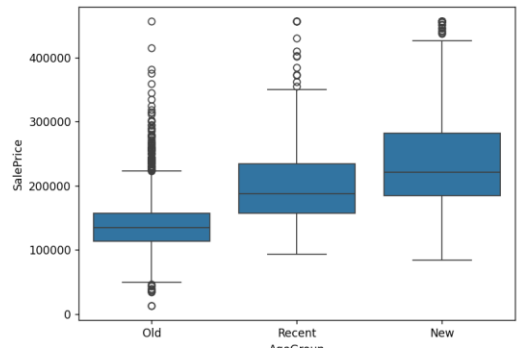
The dataset contains multiple categorical variables. Machine learning algorithms require numerical inputs, therefore LabelEncoder was applied to convert all text-based variables into integers. This final transformation ensures the dataset is fully numerical and ready for modeling.

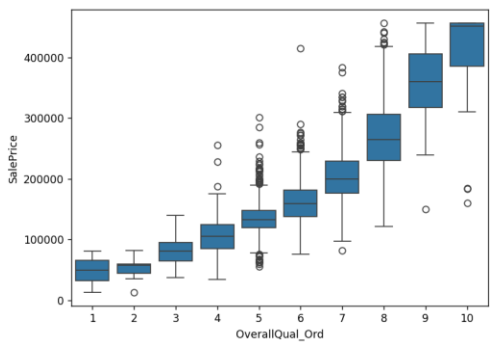
Final Output

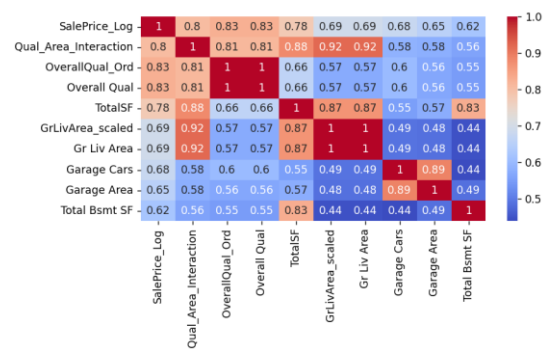
After creating new features and applying encoding, the dataset shape became:

- Rows: 2930
- Columns: 85









Conclusion

The feature engineering steps documented in this report represent a core stage in the machine learning pipeline. By creating meaningful variables and properly encoding categorical data, the dataset gains improved structure, clarity, and predictive value—leading to stronger model accuracy and overall performance.